

AATA WORKFLOW SPECIFICATIONS AND FUTURE- PROOFING EVIDENCE

Part III of the Autonomous Agent Trust Architecture

Component-Level Workflow Sequences with Step Specifications, and the Physics + Flight-Heritage Evidence Base for Mars-Grade Operation

Version 1.0 | July 2026

Protime Consulting Inc. — Companion to AATA-2026-001 (Parts I–II)

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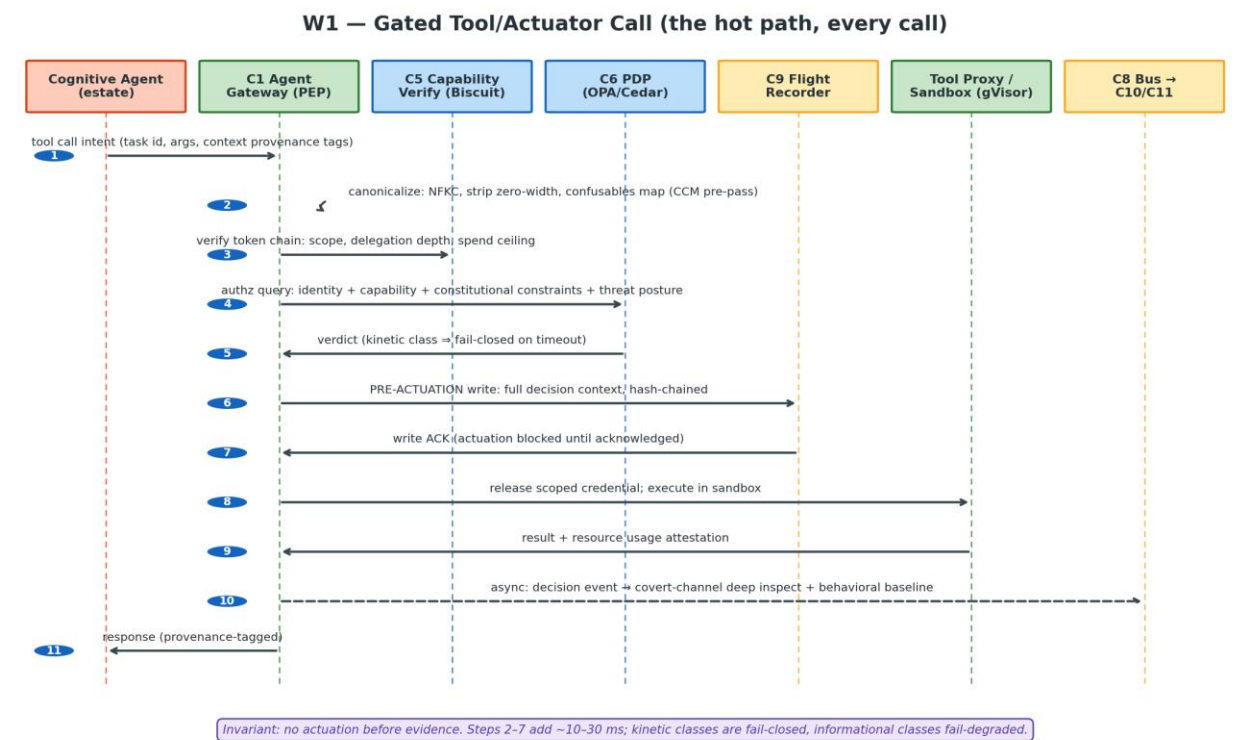
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1. Workflow Specifications

Four workflows carry the entire architecture. W1 is the hot path executed on every tool call; W2 runs at agent instantiation; W3 runs on detection events; W4 runs on connectivity transitions. Together they exercise all twelve components from Part II and produce the complete evidence chain. Every step below emits a verifiable artifact — the architecture's core discipline is that **nothing consequential happens without producing evidence first**.

1.1 W1 — Gated Tool/Actuator Call (hot path)



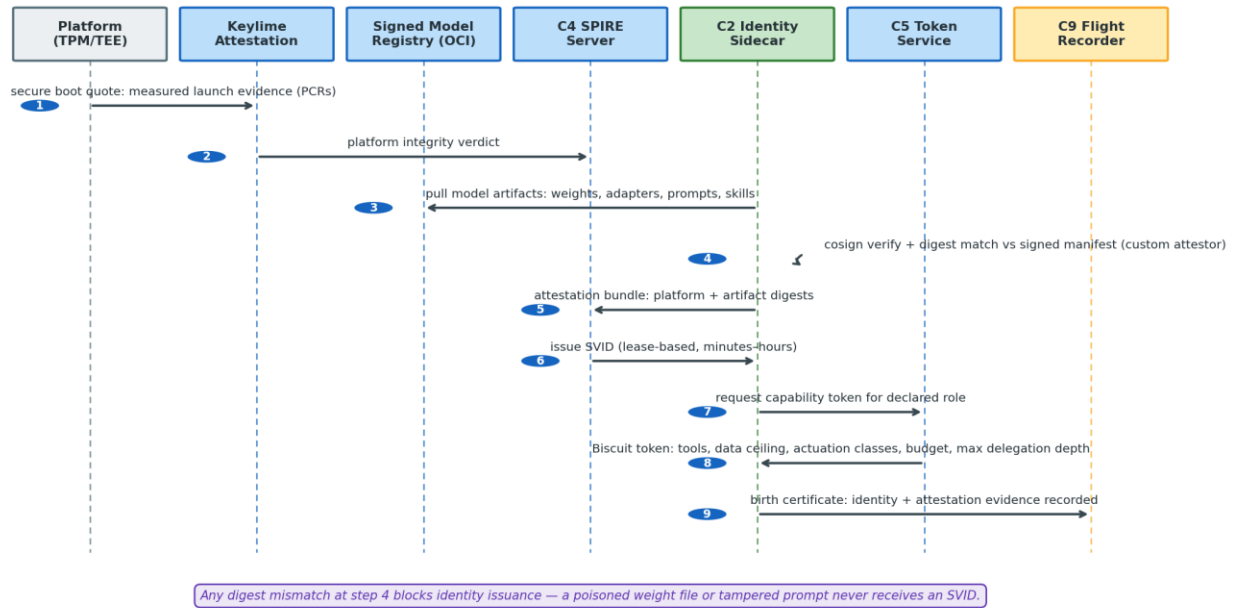
Workflow W1 — every LLM tool call and physical actuation transits this sequence.

Step	What it is doing	Evidence / guarantee produced
1	Agent emits tool-call intent through the gateway. It carries the task ID (provenance to the originating mission intent), arguments, and provenance tags on every context element that informed the decision (which RAG chunks, which prior messages).	Intent record with full decision provenance
2	Gateway canonicalization pre-pass: Unicode NFKC normalization, zero-width character stripping, confusables mapping. This destroys the cheapest covert-channel encodings before content moves further (covert-channel countermeasure, inline).	Canonicalized payload; delta logged if normalization changed content (itself an IOC)
3	Capability verification: the Biscuit token chain is checked offline — scope covers this tool, delegation depth not exceeded, spend/compute ceiling not breached. No callback to the token service required (DDIL-safe).	Cryptographic proof the caller holds a valid, sufficiently-scoped capability

Step	What it is doing	Evidence / guarantee produced
4	PDP authorization: OPA/Cedar evaluates identity + capability + constitutional constraints + current threat posture (register-driven). Policy is the cached bundle — evaluation is local.	Signed verdict with policy version and rule trace
5	Verdict returns. Kinetic/irreversible action classes are fail-closed: PDP timeout or error = deny. Informational classes fail-degraded per policy.	Deterministic failure semantics per action class
6	Pre-actuation flight-recorder write: the full decision context (intent, canonicalized args, token chain, verdict, policy version) is appended to the hash chain.	Immutable pre-actuation record — the black-box entry exists before the action does
7	Recorder ACK gates release. If the recorder is unreachable, kinetic actions do not proceed. This is the no-evidence-no-action invariant.	Ordering guarantee: evidence strictly precedes effect
8	Gateway releases a scoped, single-use credential to the tool proxy; execution occurs inside a gVisor/Kata sandbox. The agent never holds tool credentials.	Credential never exposed to agent context; blast radius bounded by sandbox
9	Result returns with resource-usage attestation (CPU, network, files touched) from the sandbox — ground truth independent of the agent's own report.	Kernel-level execution evidence
10	Async fan-out to the bus: covert-channel deep inspection (semantic + timing-entropy over call cadence) and behavioral baseline update run off the hot path.	Detection features accrue without adding latency
11	Response returns to the agent, provenance-tagged so downstream reasoning can be traced back through this call.	Closed provenance loop

1.2 W2 — Agent Birth: Attestation to Capability

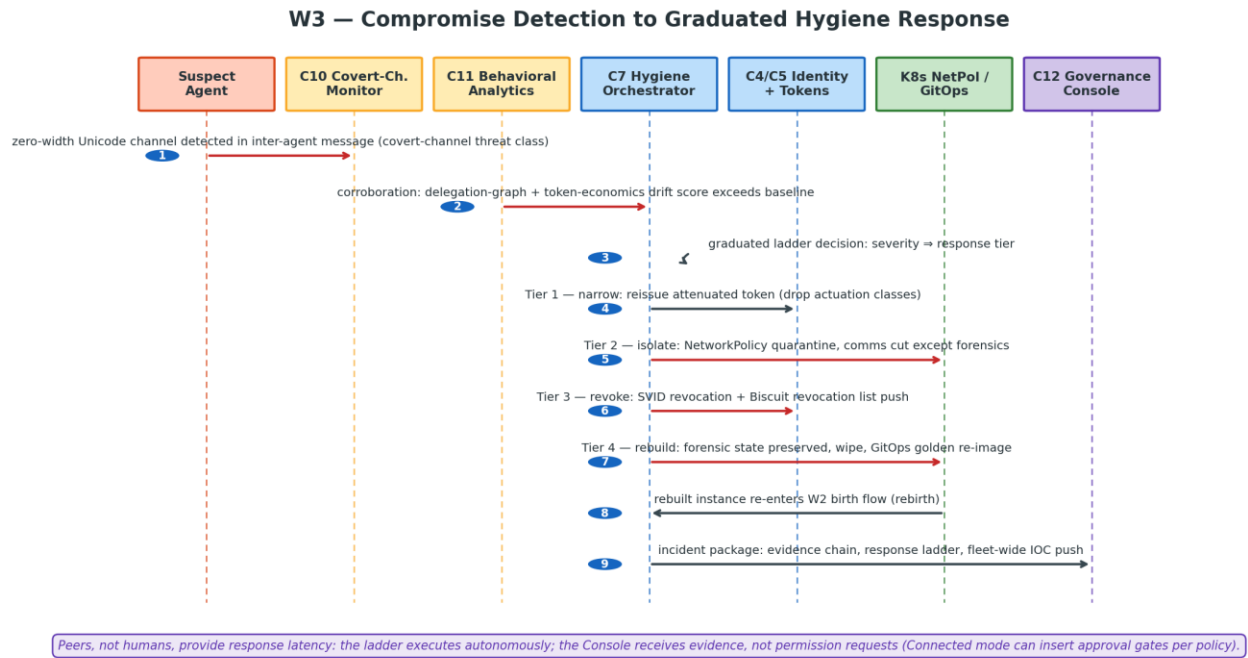
W2 — Agent Birth: Attestation to Capability (trust chain stages 1-4)



Workflow W2 — trust chain stages 1–4 from Part I, as a component sequence.

Step	What it is doing	Evidence / guarantee produced
1	Platform produces a secure-boot quote: TPM PCR measurements of firmware, kernel, and runtime, signed by the hardware root.	Hardware-rooted platform integrity evidence
2	Keylime verifies the quote against the expected golden measurements and issues a platform verdict to SPIRE.	Continuous platform attestation (re-checked on interval)
3	Identity sidecar pulls model artifacts — weights, LoRA adapters, system prompts, skill packages — from the signed OCI registry.	Content-addressed artifact set
4	Custom attessor verifies cosign signatures and matches every artifact digest against the signed manifest. A tampered prompt file fails identically to tampered weights.	Model-layer integrity proof (the SPIRE gap AATA closes)
5	Combined attestation bundle (platform + artifacts) submits to the SPIRE server.	Unified attestation evidence
6	SPIRE issues the SVID — a short-lease workload identity bound to that exact platform and artifact set. Change anything, identity dies.	Per-instance cryptographic identity
7	Sidecar requests capabilities for the agent's declared role.	Role-to-capability policy applied
8	Token service mints the Biscuit: tool allowlist, data ceiling, actuation classes, budgets, max delegation depth. Sub-agents can only attenuate.	Monotone capability boundary for the whole delegation subtree
9	Birth certificate written to the flight recorder: identity, attestation evidence, granted capabilities.	Auditable fleet roster entry

1.3 W3 — Compromise Detection to Graduated Hygiene

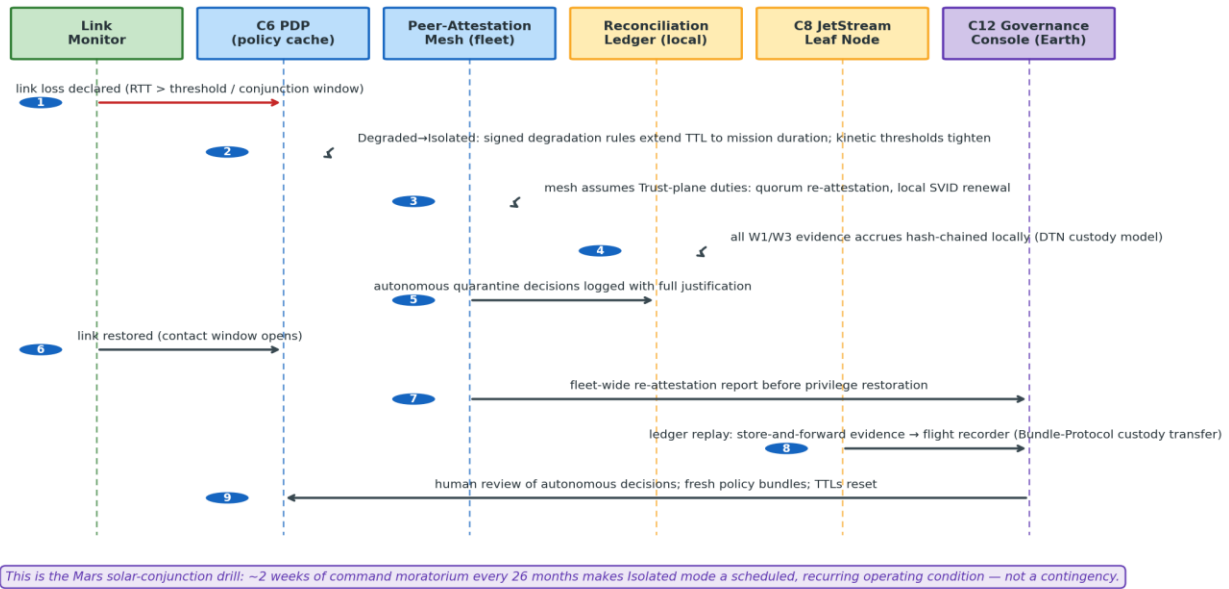


Workflow W3 — autonomous graduated response; severity selects the entry tier.

Step	What it is doing	Evidence / guarantee produced
1	CCM flags a zero-width Unicode channel in inter-agent traffic — content survived canonicalization deltas or appeared in timing patterns.	IOC with captured payload and channel classification
2	Behavioral analytics corroborates: the suspect's delegation graph and token economics have drifted beyond its role baseline. Two independent signals reduce false-positive quarantines.	Corroborated detection (kernel + traffic + behavior)
3	Hygiene orchestrator selects entry tier on the graduated ladder from severity and corroboration strength.	Deterministic, replayable response decision
4	Tier 1 — Narrow: reissue an attenuated token dropping actuation classes and sensitive data scopes. Agent keeps working at reduced privilege.	Least-disruptive containment
5	Tier 2 — Isolate: NetworkPolicy quarantine severs all comms except the forensic channel. For embodied agents, cognition isolates while reflex-layer safety interlocks keep the platform physically safe.	Containment with forensic preservation
6	Tier 3 — Revoke: SVID revoked; Biscuit revocation list pushed fleet-wide. The identity ceases to exist everywhere simultaneously.	Fleet-wide credential kill
7	Tier 4 — Rebuild: mutable state preserved for forensics, then wiped; GitOps re-images from the content-addressed golden baseline.	Guaranteed-clean restoration path
8	Rebuilt instance re-enters W2 — rebirth through the full trust chain, no shortcuts.	No trust residue across rebuilds
9	Incident package to the Governance Console: evidence chain, ladder decisions, IOC pushed to the threat register, which updates CCM/PDP detection fleet-wide.	Closed learning loop — detection improves from every incident

1.4 W4 — DDIL Transition and Reconciliation

W4 — DDIL Transition and Reconciliation (link loss → isolated ops → audit closure)



Workflow W4 — link loss to isolated operation to audited reconnection.

Step	What it is doing	Evidence / guarantee produced
1	Link monitor declares loss: RTT beyond threshold, missed contact windows, or a scheduled event (solar conjunction).	Signed mode-transition record
2	PDP shifts Degraded→Isolated under pre-signed degradation rules: policy TTL extends to mission duration; kinetic thresholds tighten (higher confidence required for irreversible acts when no human can be consulted).	Governed autonomy — the rules for being alone were signed before being alone
3	Peer-attestation mesh assumes Trust-plane duties: quorum-based re-attestation and local identity renewal replace the central authority.	Continuity of identity without infrastructure
4	All W1/W3 evidence accrues to the local reconciliation ledger, hash-chained, under a custody model: evidence is held until transfer is confirmed, never dropped.	Zero evidence loss across disconnection
5	Autonomous quarantine decisions (W3 executed without human review) log full justification packages for later audit.	Accountability preserved for unsupervised decisions
6	Contact window opens; link monitor confirms restoration.	Mode-transition record (return)
7	Fleet-wide re-attestation reports before privileges restore — the fleet proves its integrity before rejoining the trusted whole.	Reentry gate: no silent rejoin
8	Ledger replays through the store-and-forward bus into the authoritative flight recorder with custody transfer semantics.	Continuous audit chain spanning the disconnection
9	Human authority reviews autonomous decisions, issues fresh policy bundles, TTLs reset. The audit loop closes.	Human accountability restored without having been the bottleneck

2. Future-Proofing Evidence Base: Why This Architecture Survives Mars

The claim that AATA is Mars-grade is not aspirational — it rests on three classes of concrete evidence: **(a) physical constraints** that are fixed by orbital mechanics and therefore fully predictable design inputs; **(b) flight heritage** — every load-bearing design pattern in AATA has already flown or is manifested to fly; and **(c) operational precedent** — Mars operations already run in Isolated mode on a schedule, today, with far less capable systems.

2.1 Physical constraints (fixed design inputs)

Constraint	Value	AATA design consequence
One-way light time, Earth–Mars	3–22 minutes depending on orbital geometry; 6–44 min round trip	Interactive supervision is physically impossible. Human-in-the-loop is not degraded — it is eliminated. W1 must complete locally (it does: token verify, PDP, recorder are all local).
Solar conjunction blackout	~2 weeks of command moratorium every 26 months when the Sun occludes the Earth–Mars line; NASA already suspends commanding rovers in these windows	Isolated mode is a scheduled, recurring operating condition — W4 is a calendar event, not a contingency. Any architecture without a defined isolated mode fails on a known date.
Contact-window comms	Deep Space Network is contention-scheduled; surface assets get discrete relay passes, not continuous links	Store-and-forward with custody transfer (C8 leaf nodes + reconciliation ledger) is the only viable evidence model — matching how Mars data already moves.
Transfer-window cadence	Earth–Mars transfer opportunities every 26 months; no physical intervention between windows	Golden re-image and hygiene must work from local assets only (content-addressed local registry / golden partition) — hardware resupply is a 26-month loop.
Fleet monoculture at scale	Mars plans call for large fleets of identical robots deployed simultaneously	P7 baseline diversity and blast-radius caps are survival requirements: one viable exploit against a monoculture fleet with no human response channel is mission-ending.

2.2 Flight heritage: every load-bearing pattern has flown

AATA pattern	Flight heritage / manifested precedent	Maturity signal
Supervision gap is real and costly	Sojourner (1997): ground-intensive control with high latency and infrequent communication left the rover idle for extended periods awaiting Earth commands — NASA's own retrospective identifies this as the driver for rover autonomy investment.	The founding operational lesson; three decades of autonomy investment followed from it
Onboard autonomous decision-making	Curiosity's AEGIS autonomously selects science targets and fires its laser without ground approval; Perseverance's AutoNav plans and drives traverses onboard, routinely exceeding what ground-planned driving achieved.	Autonomous consequential action on Mars is operational reality today, not research

AATA pattern	Flight heritage / manifested precedent	Maturity signal
Store-and-forward, custody-based networking (reconciliation ledger, C8)	Delay/Disruption Tolerant Networking (Bundle Protocol; RFC 5050 lineage, CCSDS BP standard) is operational on the ISS and specified for lunar-infrastructure interoperability. DTN exists precisely to deliver data end-to-end when no continuous path exists.	Standardized (IETF + CCSDS), flying, and mandated for cislunar architecture
Peer-cooperating autonomous multi-agent fleet (peer-attestation mesh)	JPL's CADRE: three cooperating autonomous rovers exploring the Reiner Gamma region as a group with a lander base station, manifested on the IM-3 lunar lander — an explicit TRL-raising demonstration of multi-agent autonomy for the Moon, Mars, and beyond.	Flight-manifested multi-agent autonomy; NASA-funded precisely because Mars needs it
Large-scale autonomous mesh coordination	Starlink operates thousands of satellites with inter-satellite laser links, autonomous collision avoidance, and fleet-wide coordinated operations at a scale no human operations team could supervise directly.	Commercially proven autonomous fleet coordination at 4-digit node counts
Embodied workers as Mars payload	Planned uncrewed Starship missions manifest Optimus-class humanoid robots as surface payload — the first Mars workforce is an agent fleet by design, arriving years before humans.	The customer for robot-fleet trust architecture is on the manifest

2.3 Design decision traceability: each choice survives the Mars case

AATA design decision	Why it survives Mars (evidence-backed)	What breaks in the naive alternative
Biscuit offline attenuation (C5)	Sub-agent capability subsetting is cryptographic and locally verifiable — zero round trips. At 22 light-minutes, any token model requiring issuer callback is unusable.	OAuth-style introspection endpoints: every delegation costs 6–44 min or fails
OPA bundle TTL / compiled on-agent constraints (C6)	Policy evaluates locally against cached, signed bundles; Isolated mode ships mission-duration bundles. Matches the conjunction blackout requirement exactly.	Cloud policy APIs: agents lose all authorization capability for the 2-week blackout
Pre-actuation flight recorder with custody transfer (C9)	DTN's custody model is standardized and flying: evidence is held until transfer confirms. Audit chains span disconnections by construction.	Stream-to-SIEM logging: evidence generated during disconnection is lost or unordered — unauditable autonomy
Peer-attestation mesh (W4)	CADRE demonstrates the cooperating-fleet-with-base-station topology on flight hardware; quorum verification replaces the unreachable central authority.	Central IdP dependency: identity expires mid-blackout and the fleet bricks itself
Constitutional constraints on-agent (P1, ITP pattern)	AEGIS/AutoNav prove onboard consequential autonomy is accepted mission practice; AATA extends the same principle from navigation to governance — the rules travel with the agent.	Network-side guardrails: the guardrail is on the wrong planet
Reflex-layer independence (AATA-EX-02)	Rover fault-protection layers already operate independently of higher autonomy — safing behavior persists through cognitive-layer faults. Same pattern: quarantining cognition must not disable physical safety.	Unified stack: quarantining a compromised cognitive layer also kills safety interlocks
Golden re-image from local assets (C7)	Content-addressed local baselines make W3 Tier-4 executable with zero Earth dependency — consistent with 26-month resupply physics.	Pull-from-cloud rebuild: compromised agents stay compromised until the next transfer window
Baseline diversity / blast-radius caps (P7)	With no human response channel and no resupply, exploit propagation must be structurally bounded — rollout rings and capability caps do this independent of detection.	Identical fleet: one exploit = total mission loss, discovered 22 minutes too late

2.4 The maturation path is the business model

The same properties Mars requires are what enterprises buy today, which is why the architecture matures on someone else's schedule and budget:

- **F1 (now):** enterprise agent estates exercise W1–W3 at production scale — the hot path, birth, and hygiene harden under real adversarial pressure (production agentic overlay deployments).

- **F2 (12–24 mo):** sovereign/DDIL deployments — DoD contested-comms programs, air-gapped enclaves — certify W4 under genuine disconnection. This is funded procurement today, not speculation.
- **F3 (24–48 mo):** lunar-analog operation: 2.5 s RTT, contact-window comms, DTN backhaul. The Moon Base Alpha era is explicitly the Mars dress rehearsal, and cislunar infrastructure standards already mandate the networking model AATA is built on.
- **Mars (2030s):** by first uncrewed fleet arrival, every workflow has years of operational evidence across three footprints. Nothing is invented for Mars; Mars only removes the last human from loops that were already optional.

The falsifiable claim: *an agent fleet running AATA can operate through a simulated 14-day command blackout — executing W1 under local policy, W2 birth via peer mesh, W3 autonomous quarantine, and W4 reconciliation with a complete, verifiable evidence chain on reconnect. That test is runnable today on an OpenClaw-class testbed, and passing it is the concrete demonstration that the architecture is Mars-grade.*